Brian Lee

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PROFESSIONAL SUMMARY

Data Analyst focused on quantitative analysis, statistical modeling, and advanced data processing with Python/MATLAB. Solid background in data automation, model development, and visualization, with interdisciplinary collaboration experience. Skilled in exploratory data analysis, algorithm development, and communicating results to technical audiences.

EXPERIENCE

Research Data Analyst — **Arlington, Virginia**
Virginia Tech National Security Institute — *Aug. 2023 - Aug. 2025*

- **Systems integration:** Integrated computer vision models into **MIMIK**, a mission engineering platform for simulating military kill webs, to enable performance analyses of AI-enabled components within system-of-systems architectures.
- **Research software for interdisciplinary application:** Led the development of CODEX, a Python-based framework that applies combinatorial testing to analyze machine-learning datasets and characterize operating envelopes for AI-enabled systems.
- **Interdisciplinary application:** Applied CODEX across **5+** operational domains in RF, Bayesian test modeling, and NLP to evaluate system performance and robustness, under data-constrained conditions.
- **Community engagement:** Designed and delivered paper presentations and live demonstrations to **60+** national security T&E researchers through CAI 2025, DataWorks 2025, and the 2024 CT4AIES Workshop.

Research Intern — **Arlington, Virginia**
Virginia Tech National Security Institute — *May 2022 - Aug. 2022*

- **Automated dataset curation:** Constructed an image data pipeline in Python that scrapes, preprocesses, and filters Instagram images by hashtag, assembling **30+** curated datasets for training GANs from **50,000+** images.
- **Model training:** Systematically trained **55+** NVIDIA *StyleGAN3* models using GPU/CUDA-accelerated workflows, applying transfer learning to achieve FID score convergence (stable image quality) **1.5**× faster under experimental conditions.

PROJECTS

Conway's Game of Life — *Nov. 2021 - Dec. 2021*

- Implemented and executed large-scale simulations of John Conway's *Game of Life* on an HPC cluster using parallel computing to update **10 billion+** states, scaling to 128 MPI tasks with batch-scheduled execution.
- Conducted quantitative strong scaling tests to measure a **20%** speedup under parallel execution.

Dynamics of a Hierarchal Predator-Prey System — *Apr. 2022 - May 2023*

- Developed a nonlinear population dynamics model for a hierarchical predator–prey system using coupled ODEs.
- Simulated and analyzed the existence of nontrivial equilibrium states corresponding to stable population coexistence and extinction, using parameter sweeps in MATLAB to characterize oscillatory behavior and stability transitions.

PUBLICATIONS

- Lanus, E., **B. Lee**, et al., "Coverage for Identifying Critical Metadata in Machine Learning Operating Envelopes." *2024 IEEE International Conference on Software Testing, Verification, and Validation Workshops*. Toronto, ON, CA, 2024.
- S. Mhatre, **B. Lee**, et al., "Coverage Metrics for Detecting Bias in Facial Recognition Datasets," in 2025 IEEE Conference on Artificial Intelligence (CAI), May 2025.

EDUCATION

Virginia Tech — **Blacksburg, Virginia**
B.S. in Computational Modeling and Data Analytics — *Aug. 2019 - May 2023*
Minors in Computer Science and Mathematics — GPA: **3.48/4**

Technical Skills: Python, C/C++, MATLAB | MPI, Linux/UNIX HPC, Git | Numerical Methods for PDEs and Dynamical Systems | Model Validation and Uncertainty Quantification